Integrated Operation Enhancement: Resolving Processing Delays and Improving Inventory Management

A Mid-Term report for the Business Data Management Capstone Project

Submitted by

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Executive Summary

Title: Integrated Operation Enhancement: Resolving Processing Delays and Improving Inventory Management

Balaji Powder Coating, led by **Mr. Arup Bera**, is an industrial powder-coating facility located in **Lakshmanpur Industrial Estate, Domjur, Howrah**. The company provides advanced surface finishing for metal components such as aluminium frames, steel furniture, beams, and battery boxes, catering to clients in the electrical, fabrication, and construction sectors. Despite its technical expertise, the factory faces recurring operational challenges — primarily **inventory mismanagement and production delays**. These issues are worsened during **monsoon seasons**, when **rainwater flooding in the industrial area** disrupts material movement, storage, and scheduling, resulting in delayed deliveries and inefficient resource use.

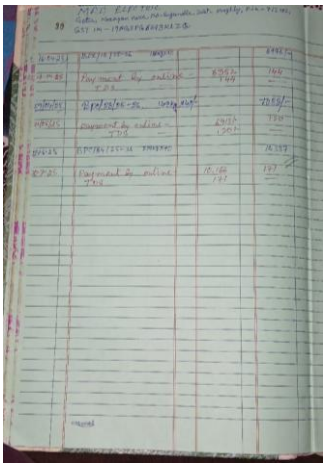
For this study, **data was collected from production logs, job cards, and dispatch registers**, along with a **survey of employees** to capture ground-level operational constraints. The dataset includes fields such as **Item Serial No, Item Name, Quantity, Price, Order Date, Delivery Date and Customer Name.** Descriptive statistics revealed frequent mismatches between available and recorded inventory, and significant downtime during periods of heavy rainfall.

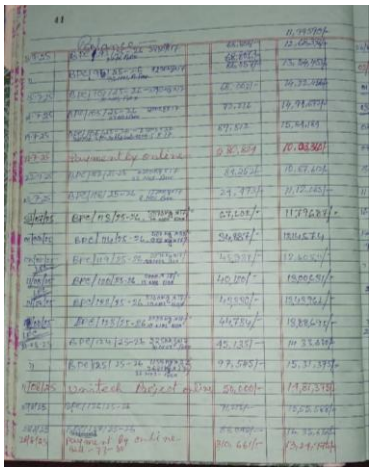
The analysis emphasizes **inventory streamlining and delay minimization** through workflow time evaluation and digital record-keeping. A **data-backed inventory management system** was developed using **Microsoft Excel**, chosen due to the company's **limited IT infrastructure**, absence of specialized software, and the **need for an easily transferable, low-maintenance solution**. Excel provides **flexibility in data entry, compatibility with existing records, and minimal training requirements** for staff. Mid-term results indicate **improved stock accuracy and reduced process delays**, highlighting the practicality and effectiveness of this data-driven intervention in enhancing operational efficiency.

Proof of Data Originality

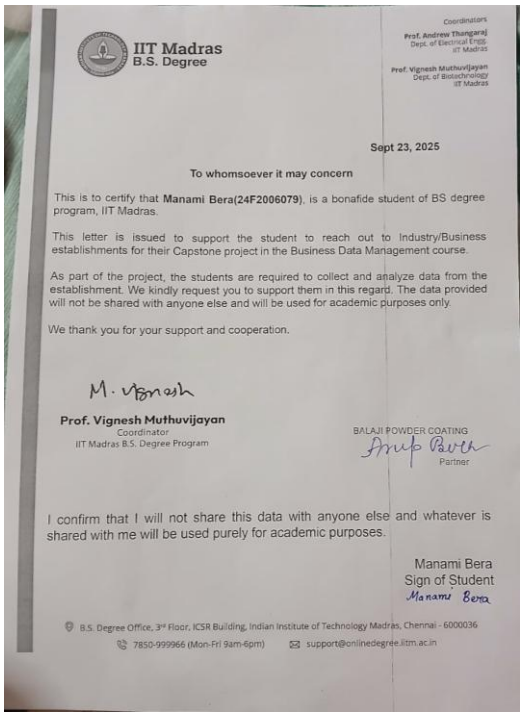
Here I am providing a sample of data I got from the company. I am adding the links of the Authorization letter, company and factory pictures. I am also adding a video showing my conversation with the Owner of the Factory and 2 sample pictures of the register copy with the data.

Picture of Register with the Data:





Authorization Letter:



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Sept 23, 2025

To whomsoever it may concern

This is to certify that **Manami Bera(24F2006079)**, is a bonafide student of BS degree program, IIT Madras.

This letter is issued to support the student to reach out to Industry/Business establishments for their Capstone project in the Business Data Management course.

As part of the project, the students are required to collect and analyze data from the establishment. We kindly request you to support them in this regard. The data provided will not be shared with anyone else and will be used for academic purposes only.

We thank you for your support and cooperation.

M. Vignesh

Prof. Vignesh Muthuvijayan
Coordinator
IIT Madras B.S. Degree Program

BALAJI POWDER COATING
Arup Pavth
Partner

I confirm that I will not share this data with anyone else and whatever is shared with me will be used purely for academic purposes.

Manami Bera
Sign of Student
Manami Bera

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Photo of the Factory:

Link of the Folder which contains all pictures and Videos: BDM Folder

Link of the Communication between Me and Mr. Arup: Communication Video

Link of the Excel Worksheet: Data_BDM

Meta Data and Descriptive Analysis

Meta Data

Data Collection Period	Data is collected from April 2025 to September 2025.
Mode of Data Collection	I have collected the data from the factory's manually maintained registers and systematically entered it into an Excel file for analysis.
Details of the Data Provided	Data containing the details like Item Name, Customer Name, Order Date, Quantity, Price, etc.

Data Received:

The Data I got consists of all the information like Customer Name, Order Date, Total number of Items and Prices for powder coating them all. The data is within the duration of April 2025 to September 2025. This data will be used for the analysis.

Serial No.	Item Name	Quantity	Price	Order Date	Customer Name
BPC/01/25-26	Battery Box	32	76629	28-03-2025	Unitech Projects
BPC/02/25-26	Metal Rack	120	18861	28-03-2025	Bholanath Powder Coating
BPC/04/25-26	Beam	266	46640	01-04-2025	Steelwyn Corporation
BPC/13/25-26	Aluminium Frame	30	6726	12-04-2025	Chintamani Steel Metal Works
BPC/18/25-26	Hospital Bed	17	98093	17-04-2025	Unitech Projects

Table-1 is a sample copy of the data received from the factory owner.

Descriptive Statistics

April 2025

- There were 23 orders in April.
- Total Revenue generated is Rs 11,36,214.
- Battery Box generated the most revenue i.e. Rs 2,86,575, by powder coating 440 units.
- Beam is the most powder coated item (1083 units powder coated) generating Rs 2,11,250

May 2025

- There were 29 orders in May.
- Total Revenue generated is Rs 10,51,572.
- Hospital Beds generated the most revenue i.e., Rs 4,24,187, by powder coating 132 units.
- Metal Rack is the most powder coated item (2244 units powder coated) and generated Rs 3,07,42.

June 2025

- There were 24 orders in June.
- Total Revenue generated is Rs 10,52,593.
- Box generated the most revenue i.e., 3,04,335, by powder coating 115 units.
- Battery Box is the most powder coated item (1442 units powder coated) and generated Rs 1,35,228.

July 2025

- There were 24 orders in July.
- Total Revenue generated is Rs 11,51,407.
- Box generated the most revenue i.e., 3,85,352, by powder coating 124 units.
- Steel Furniture is the most powder coated item (230 units powder coated) and generated Rs 2,66,757.

August 2025

- There were 15 orders in August.
- Total Revenue generated is Rs 8,18,921.
- Box generated the most revenue i.e., 6,45,463, by powder coating 265 units.
- Box is the most powder coated item (265 units powder coated) and generated Rs 6,45,463.

September 2025

- There were 26 orders in September.
- Total Revenue generated is Rs 8,60,235.
- Box generated the most revenue i.e., 3,69,694, by powder coating 165 units.
- Box is the most powder coated item (165 units powder coated) and generated Rs 3,69,694.

Detailed Explanation of Analysis Process

Data Cleaning and Preprocessing

The raw data for this study was **collected directly from the factory's manually maintained production and dispatch registers**, which recorded daily order details, material inflow, and completion timelines. I have collected the data from these registers and systematically entered it into an **Excel file** for analysis.

During **data cleaning**, redundant entries, incomplete records, and inconsistencies (e.g., missing dispatch dates or duplicate Order IDs) were identified and corrected. Null values were replaced through logical estimation based on surrounding data trends (e.g., average processing time for the same product type). Typographical errors in product names and dates were standardized for consistency.

Importance:
Data cleaning ensured that subsequent analyses were **accurate and unbiased**, enabling reliable insights into process delays and inventory mismatches. Clean, structured data is

essential for producing meaningful graphical representations, Pareto charts, and summary statistics.

Analysis Process / Method

The analytical framework follows a structured, step-wise approach:

1. **Data Selection and Categorization:**
   Orders were categorized based on **product type, processing time, and dispatch date** to understand workflow behaviour and resource utilization patterns.

2. **Descriptive Statistical Analysis:**
   Basic statistical measures such as **mean processing time, average dispatch delay, and frequency of delayed orders** were computed. These metrics established the baseline operational efficiency.

3. **Trend and Delay Analysis:**
   Weekly processing times were plotted to visualize fluctuations in performance.

   - *Figure 1* depicts the **processing delay trend**, with **Week 35** showing the highest count of delays.
   - This spike was attributed to **heavy rainfall and flooding** in the industrial area, which disrupted production and material handling.

4. **Inventory Analysis using Pareto Principle (80/20 Rule):**
   Pareto charts were constructed to identify the **most frequently delayed or stock-out-prone items**.

   - This helped pinpoint **20% of product types responsible for 80% of the delays**, enabling targeted corrective measures.
   - *Equation:*

$$P_i = \frac{f_i}{\sum f_i} \times 100$$

   where P_i represents the percentage contribution of item *i* to total delays.

5. **Workflow Time Mapping:**
   A **process timeline chart** was developed to visualize order flow from receipt to dispatch. Bottlenecks were identified at material preparation and coating stages due to untracked intermediate inventory.

6. **Implementation of Data-Driven Inventory Model:**
   Based on insights from the analyses, a **structured Excel-based inventory tracker** was designed. It records material inflow, usage, and balance in real time. Excel was chosen for its **low cost, easy accessibility, and alignment with the company's limited IT infrastructure**, making it easy for staff to adopt and maintain.

SWOT Analysis

A **SWOT Analysis** was conducted to assess Balaji Powder Coating's internal strengths and weaknesses, along with external opportunities and threats, based on operational data and observed challenges.

Strengths

- **Experienced Leadership:** Guided by **Mr. Arup Bera**, the factory has a skilled team ensuring consistent product quality.
- **Diverse Client Base and Product Range:** Serves multiple industries—electrical, construction, etc. with varied metal products, ensuring business stability.
- **In-House Operations:** Complete powder-coating within the facility enables better quality control and quicker turnaround.

Weaknesses

- **Manual Record Management:** Reliance on handwritten registers causes errors, duplication, and time delays.
- **Poor Inventory Tracking:** No real-time visibility leads to stock mismatches and production halts.
- **Flood-Related Disruptions:** The low-lying location faces **rainwater flooding** during monsoon, delaying operations.

Opportunities

- **Excel-Based Digital System:** A cost-effective digital solution can streamline inventory and scheduling without major infrastructure changes.
- **Data-Driven Process Improvement:** Trend and Pareto analyses can identify key bottlenecks and reduce idle time.

Threats

- **Seasonal Flooding and Power Interruptions.**
- **Rising Raw Material Costs and Competitive Pricing Pressure.**

Strategic Insight

The analysis reveals that while the company's technical expertise and client trust are strong assets, **manual systems and environmental risks** hinder efficiency. Adopting **Excel-based digital tracking** and using **data insights for workflow optimization** can help mitigate delays and strengthen overall operational resilience.

Rationale and Justification

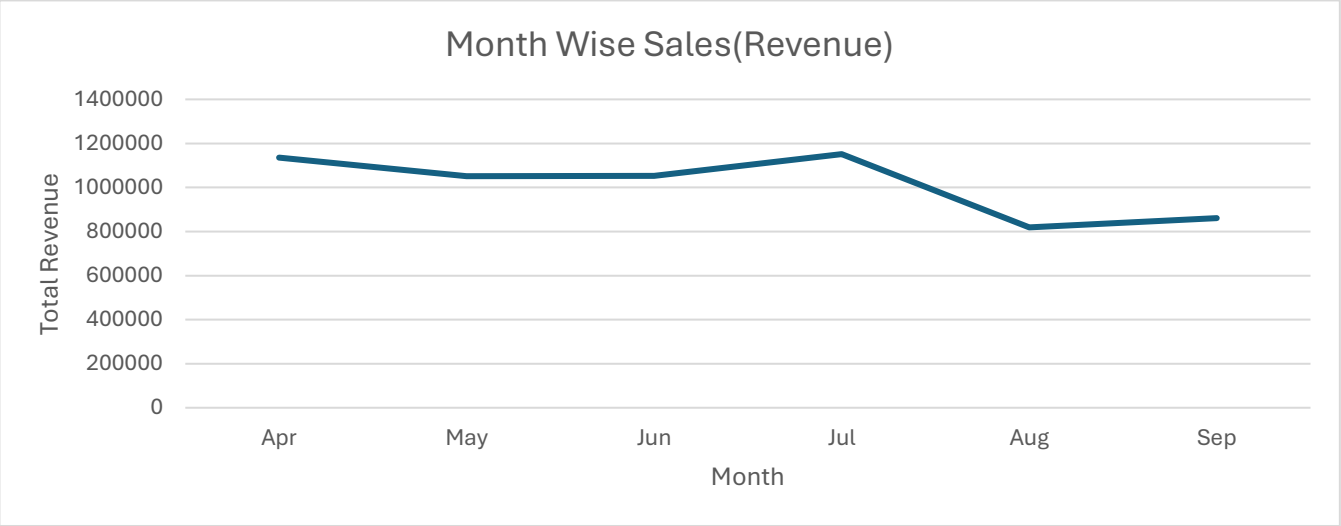
The chosen methods—**Trend Analysis, Pareto Analysis, and SWOT analysis**—directly align with the company's core challenges: **inventory management and process delays**. Trend analysis helps track performance over time, Pareto analysis prioritizes critical problem areas, and SWOT contextualizes operational risks. Compared to advanced statistical tools, **Excel-**

based methods are practical and sustainable for Balaji Powder Coating's current operational setup, ensuring ease of handover and long-term usability.

Results and Findings

Results and Findings of April 2025 to September 2025 have been shown below.

Monthly Revenue Trend Analysis



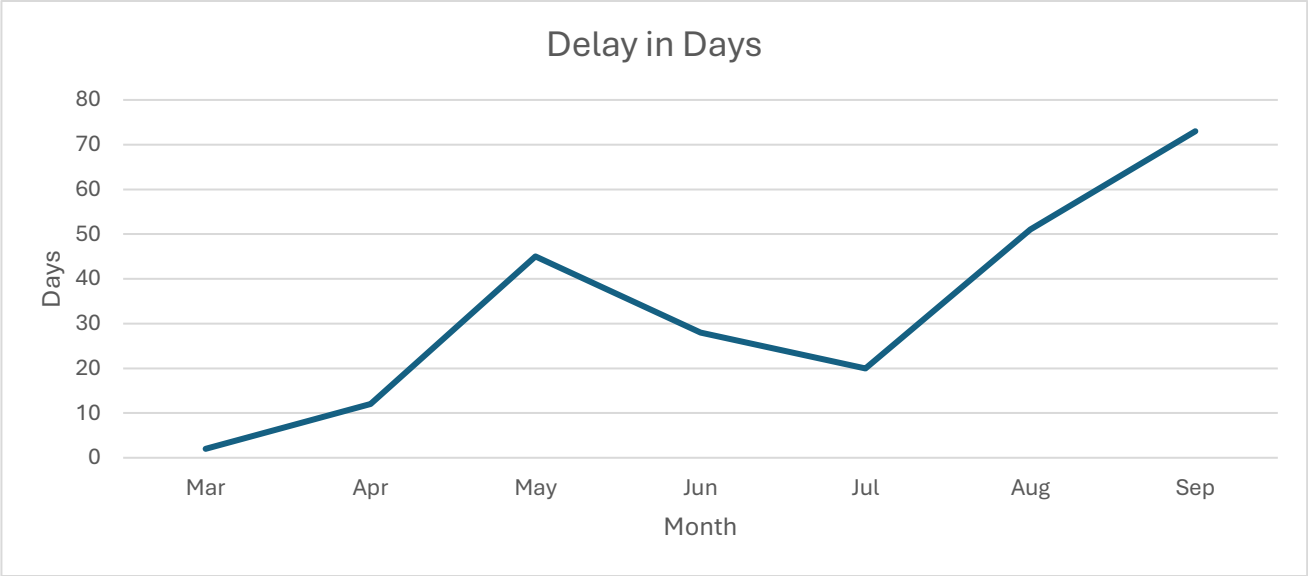
In the above graph, we see a noticeable decline in revenue can be observed during the months of **August and September** (more than 10lakhs of revenue is generated per month in April to July but an average of only 8.1lakhs in August and September), primarily due to **seasonal rainfall and flooding** in the area. These weather conditions force the factory to **halt operations frequently**, leading to prolonged shutdowns throughout the monsoon period.

Monthly Order Trend Analysis

In the above graph, we can clearly see that the **number of orders per month drops significantly in August**, corresponding to the period of heavy rainfall and flooding that disrupts production. However, there is a **gradual rise in orders mainly toward the end of September with few**

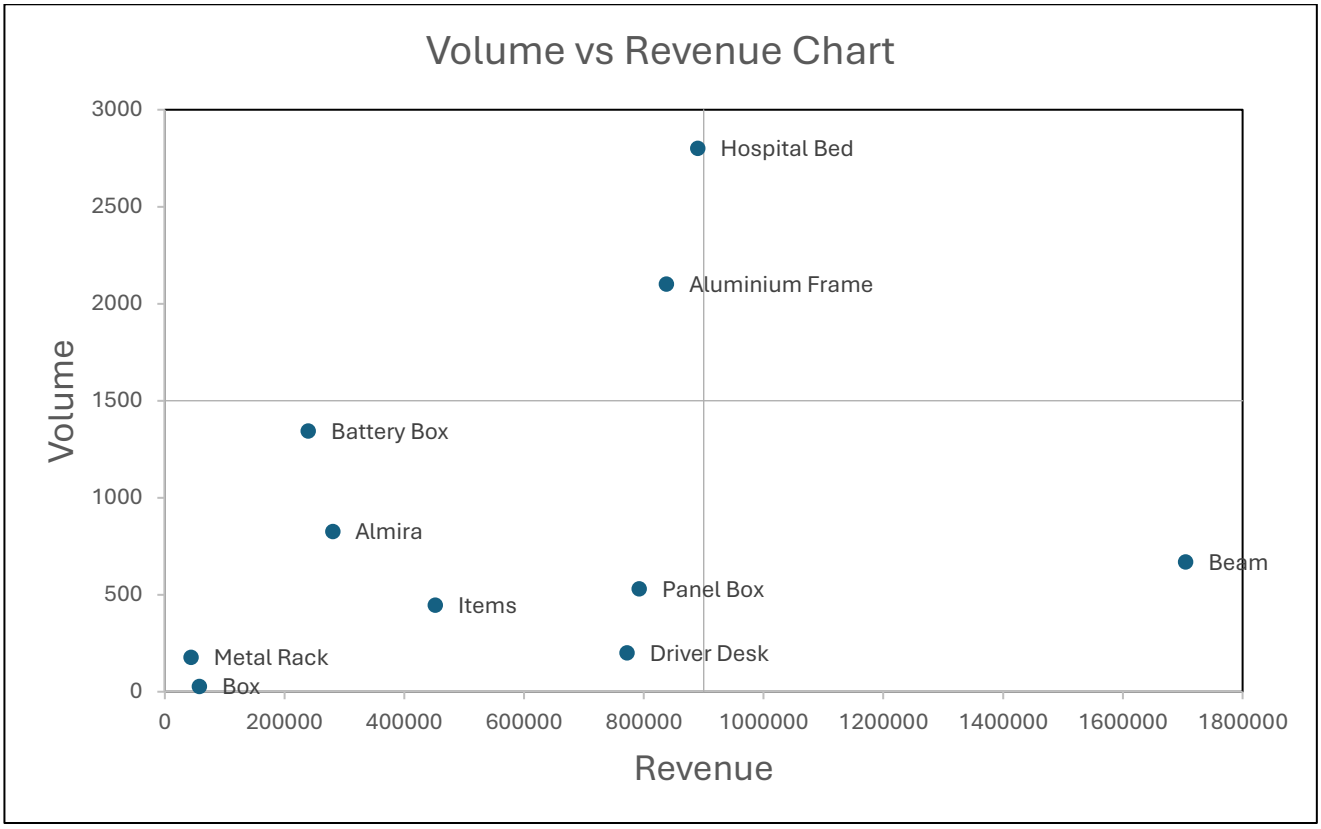
smaller orders in between, as operations resume and clients begin placing new orders once the weather conditions improve.

Delay in Processing Time Due to Seasonal Rainfall



The above graph shows a steady rise in delays from March to September, with the highest delay of around 70 days in September. While operations ran smoothly until May, the situation worsened from August onward due to heavy rains and flooding in the industrial area, which disrupted transportation and halted production. The data highlights the strong impact of seasonal weather on delivery timelines, emphasizing the need for advance scheduling and better inventory management before the monsoon period.

Inventory Data Management



The above scatter plot represents the relationship between Revenue (X-axis) and Volume or Quantity (Y-axis) for different products. This visualization helps identify which items contribute more to total revenue and which ones occupy a large portion of the inventory.

- Hospital Beds and Aluminium Frames have high volume but moderate revenue, indicating possible overstocking or slow-moving inventory.
- Beams, on the other hand, generate high revenue with relatively lower volume, suggesting efficient inventory turnover.
- Items like Metal Rack Boxes and Driver Desks show low volume and low revenue, and can be monitored for optimization or phased out if they do not add significant value.

By analysing this distribution, the company can prioritize high-revenue and fast-moving items, while reducing excess inventory in low-performing categories to improve storage and working capital efficiency.

To Summarise from the above Results and Findings Section…

Issue:

The current inventory system shows an imbalance between product volume and revenue contribution. Certain products, such as *Hospital Beds* and *Aluminium Frames*, occupy significant inventory space but generate relatively lower revenue, indicating overstocking or slow-moving items. This leads to increased holding costs and inefficiencies in resource allocation.

Solution:

Implement a data-driven inventory optimization strategy using **Pareto and Volume–Revenue analysis**. Focus on maintaining optimal stock levels for high-revenue, fast-moving items like *Beams*, while reducing or restructuring stock for low-performing products. Regular monthly reviews and forecasting models can help align inventory with actual demand, reducing wastage and improving overall profitability.